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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

Past Paper · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 60 / 60 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Functional-Skills-Maths-Level-2-Past-Paper-QP-A

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Examiner use only

Activity 1: Shopping centre

1 (a) Jessie is the manager of a shopping centre.

She knows that:

- $\frac{3}{5}$ of the shops sell clothes
- $\frac{1}{8}$ of the shops sell electronic goods
- all of the other shops sell food.

What fraction of the shops sell food?

[2 marks]

Q1(a) Worked Solution:

11
40
Award 2 marks if correct answer given
1
24
40
and 5
40
seen OR
29
[2 marks]

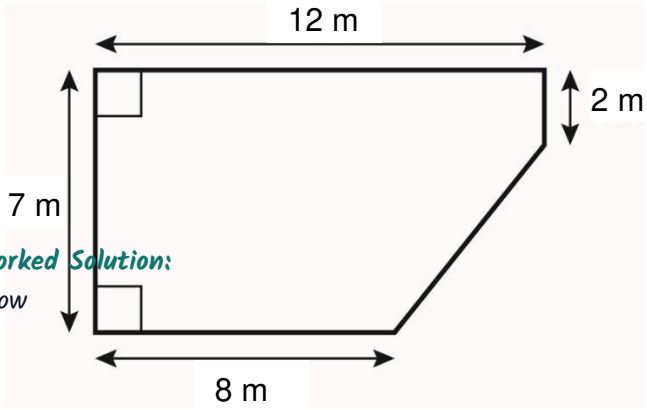
Your answer:

PAST PAPER

Please turn over

Examiner use only

1 (b) Jessie will let this unit in the shopping centre:



Not drawn accurately

Q1(b) Worked Solution:
See below
1

(7 – 2) × (12 – 8) × 0.5 or 10

OR She has information about the floor areas of other available units in the shopping centre:

(8 + 12) × 0.5 × (7 – 2) or 50

Unit floor area (m^2)	51.1	82.3	67.6	49.8	71.2	40.7	59.4	68.9	90.2

Jessie thinks that the unit she will let has a floor area 12% larger than the median floor area of other available units.

OE any method to work out area of

Is she correct?

Show how you decide.

[5 marks]

triangle or trapezium
1
74 (m^2)
Implies 1st mark
[5 marks]

Your answer:

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Examiner use only

1 (c) Jessie lets a unit with a floor area of 40 m^2 for 12 months.

The unit costs £27.59 per m^2 per month.

What is the total cost of letting this unit for 12 months?

[2 marks]

Q1(c) Worked Solution:
(£)13243.20
Award 2 marks if correct answer given
S
Level 2 – P001491 v1.5
1
27.59 × 40 × 12 or 13243.2
1
(£)13243.20
2 dp required
[2 marks]

Your answer: £

PAST PAPER

Please turn over

Examiner use only

1 (d) Jessie needs to work out how much bonus she will pay each of her workers for the last month.

She uses this formula to work out the bonus payment: $B = \frac{0.015T}{N}$

Where: B = bonus payment per worker (£)
T = total sales (£)
N = number of workers

Last month the total sales were £38 000

Jessie had 3 workers.

Work out the bonus payment each worker will get.

[3 marks]

Q1(d) Worked Solution:

(£)190
Award 3 marks if correct answer given
1
(0.015 × 38000) ÷ 3
OE any full correct method
1
570
Implied by 190
1
(£)190
[3 marks]

Your answer: £

PAST PAPER

Examiner use only

1 (e) Jessie writes a report about the number of shoppers at the shopping centre.

She writes,

"Last week, we were only open for 6 days because of a bank holiday. The mean number of shoppers was 21 thousand a day and the range was 19.5 thousand shoppers."

Jessie has this information about this week:

Q1(e) Worked Solution:

See below
1
23.2 or 23 200
H2S
1
27.1 or 27 100
H2S
1

Day of the week	Shoppers this week (in thousands)
Monday	15.4
Tuesday	12.5
Wednesday	17.3
Thursday	19.8
Friday	26.2
Saturday	39.6
Sunday	31.6

Valid comparison of both mean and range

Compare this week and last week using the means and the ranges.

[3 marks]

e.g. the mean number of shoppers this

[3 marks]

Your answer:

[Total marks: 15]

This is the end of Section A.

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Examiner use only

Activity 2: Water savings

2 (a) Kamrul wants to reduce the amount of water he uses at home.

He uses the kitchen tap for 350 days a year for 20 minutes each day.

The flow rate of his kitchen tap is 5.5 litres per minute.

Kamrul wants a new kitchen tap with a flow rate of 4.7 litres per minute.

Q2(a) Worked Solution:

He thinks he can save more than 6000 litres of water in a year if he uses the new kitchen tap for 350 days a year for 20 minutes each day.

See below

1 Is Kamrul correct?

Show how you decide.

[2 marks]

(5.5 – 4.7) × 20 × 350 or
(5.5 × 20 × 350) – (4.7 × 20 × 350) or
38500 – 32900 or 5600
OR
6000 ÷ 350 or 17.14.. AND (5.5 – 4.7) × 20 or 16
1
No AND 5600 or
[2 marks]

Your answer:

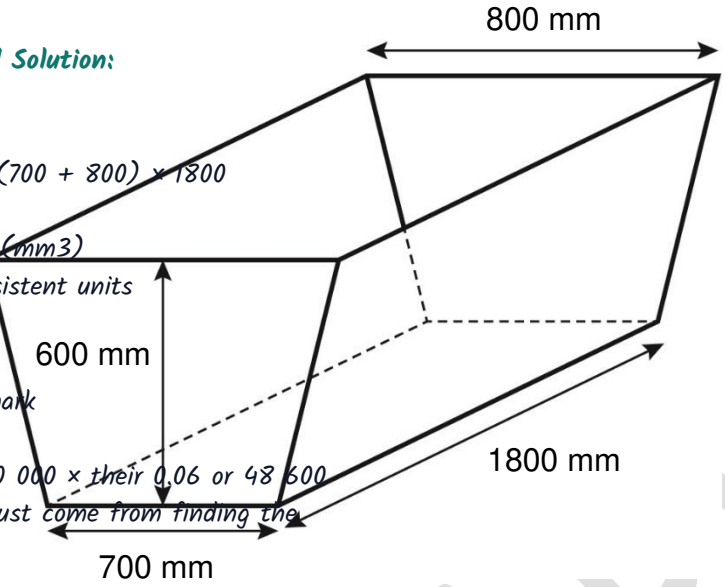
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Please turn over

Examiner use only

2 (b) The diagram shows Kamrul's bathtub.

The bathtub is a prism with a cross section in the shape of a trapezium.



Not drawn accurately

Q2(b) Worked Solution:

See below

I

0.5 × 600 × (700 + 800) × 1800

I

810 000 000 (mm3)

OE with consistent units

Implies 1st mark

I

Their 810 000 000 × their 0.06 or 48 600

Their 0.06 must come from finding the

[5 marks]

Kamrul uses this formula to calculate the capacity of this prism:

$$C = 0.5h(a + b)L$$

Where:

C = capacity of the prism
a = width of the bottom edge of the trapezium
b = width of the top edge of the trapezium
h = height of the prism
L = length of the prism

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Examiner use only

Kamrul normally fills the bathtub with water to 78% of its capacity.

He thinks that, if he fills the bathtub to 72% of its capacity, he will save more than 40 litres of water.

Is Kamrul correct?

Show how you decide.

[5 marks]

Your answer:

PAST PAPER

Please turn over

Examiner use only

2 (c) Kamrul wants to wash his car.

He knows it takes 120 litres of water to wash a car in a car wash.

Kamrul decides to hand wash his car.

He fills up a 3-gallon bucket six times to wash his car.

Q2(c) Worked Solution:

What percentage of 120 litres does Kamrul save?

Use the conversion: 1 gallon = 4.55 litres

[3 marks]

31.75(%)
Award 3 marks if correct answer given
1
3 × 4.55 or 13.65 or
3 × 6 × 4.55 or 81.9 or
120 ÷ 4.55 or 26.37..
OE any correct conversion
1
[120 – (3 × 6 × 4.55)] ÷ 120 × 100 or
(3 × 6) ÷ (120 ÷ 4.55) × 100
[3 marks]

Your answer: %

PAST PAPER

Examiner use only

2 (d) Kamrul looks at his quarterly water bill.

Total volume of water used this quarter: 118 m^3

This is 23% less than the same quarter last year.

Q2(d) Worked Solution:

Work out the volume of water Kamrul used for the same quarter last year.

Give your answer to 1 decimal place.

[2 marks]

153.2 (m3)
Award 2 marks if correct answer given
1
118 ÷ 0.77 or 153(.24..)
1
153.2 (m3)
[2 marks]

Your answer:	**m^3**

PAST PAPER

Please turn over

Examiner use only

2 (e) The water company needs to reduce water leakage.

This data shows the volumes of water leaked in 2019:

Water leaked (millions of litres per day)	Number of days
0 ≤ w < 200	87
200 ≤ w < 400	105
400 ≤ w < 600	153
600 ≤ w < 800	20
Total	365

Q2(e) Worked Solution:

Use the data to work out an estimate of the mean amount of water leaked per day.

358 (millions of litres per day)

Give your answer to the nearest whole number.

[3 marks]

Award 3 marks if correct answer given

1

(87 × 100) + (105 × 300) + (153 × 500) + (20 × 700) or

8700 + 31500 + 76500 + 14000 or

130700

Must use midpoints

H24

1

[3 marks]

Your answer: **millions of litres per day**

[Total marks: 15]

PAST PAPER

Examiner use only

Activity 3: Music festival

3 (a) Pavel organises a music festival.

He puts this information on the festival's website:

"Four million, nine hundred and eight thousand, six hundred and thirty people attended our festival over the last 30 years".

Write four million, nine hundred and eight thousand, six hundred and thirty as a number.

[1 mark]

Q3(a) Worked Solution:

4 908 630

Accept commas or spaces

[1 mark]

Your answer:

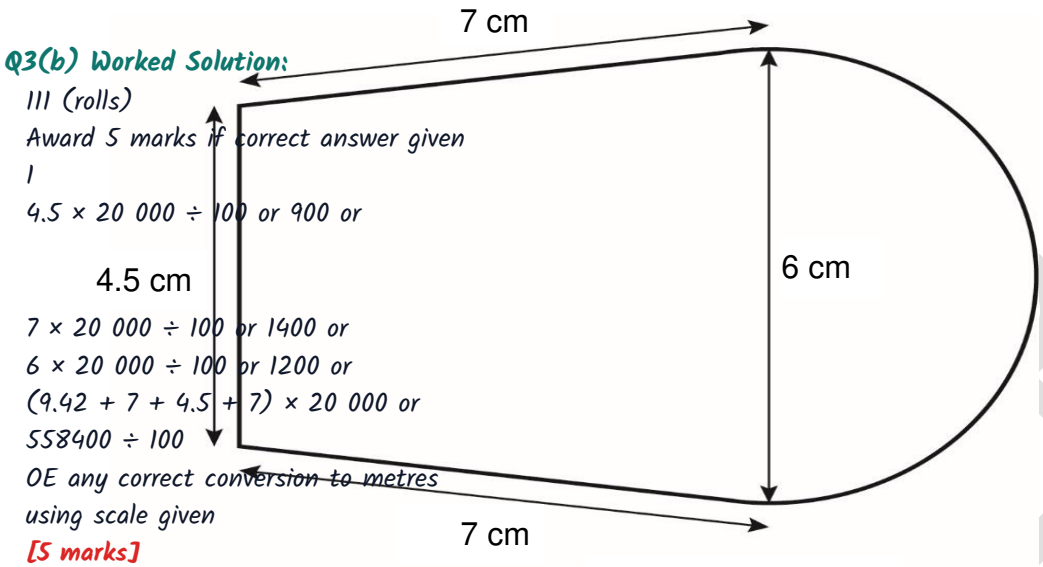
PAST PAPER

Please turn over

Examiner use only

3 (b) Pavel has this scale drawing of the festival area.

It is made up of a trapezium and a semicircle.



Scale 1 : 20 000

Q3(b) Worked Solution:
111 (rolls)
Award 5 marks if correct answer given
1
4.5 × 20 000 ÷ 100 or 900 or
7 × 20 000 ÷ 100 or 1400 or
6 × 20 000 ÷ 100 or 1200 or
(9.42 + 7 + 4.5 + 7) × 20 000 or
558400 ÷ 100
OE any correct conversion to metres using scale given
[5 marks]

He needs to order fencing to go around the festival area.

He will leave gaps totalling 65 m for the gates.

Fencing comes in rolls 50 m in length.

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Examiner use only

How many rolls does Pavel need for the festival area?

Use $\pi = 3.14$

[5 marks]

Your answer: **rolls**

PAST PAPER

Please turn over

Examiner use only

3 (c) The main stage is a square-based pyramid.

Pavel needs to work out the surface area of the four triangular faces of the pyramid.

Q3(c) Worked Solution:

No and 2400 (m2)

Award 2 marks if correct answer given

I

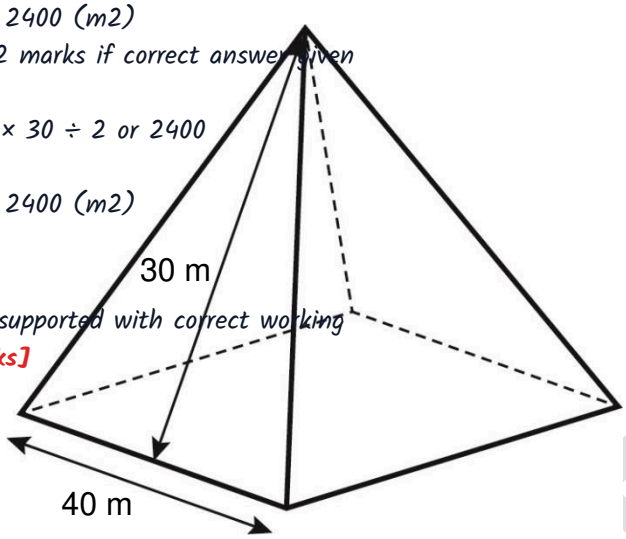
4 × 40 × 30 ÷ 2 or 2400

I

No and 2400 (m2)

OE No supported with correct working

[2 marks]



Not drawn accurately

He uses this formula: $A = 4\left(\frac{bh}{2}\right)$

Where:

A = surface area of the 4 triangular faces of the pyramid (m^2)

b = length of the base of the triangular face (m)

h = slant height of the triangular face (m)

Pavel thinks the total surface area of the 4 triangular faces of this pyramid is 1200 m^2

Is Pavel correct?

Show how you decide.

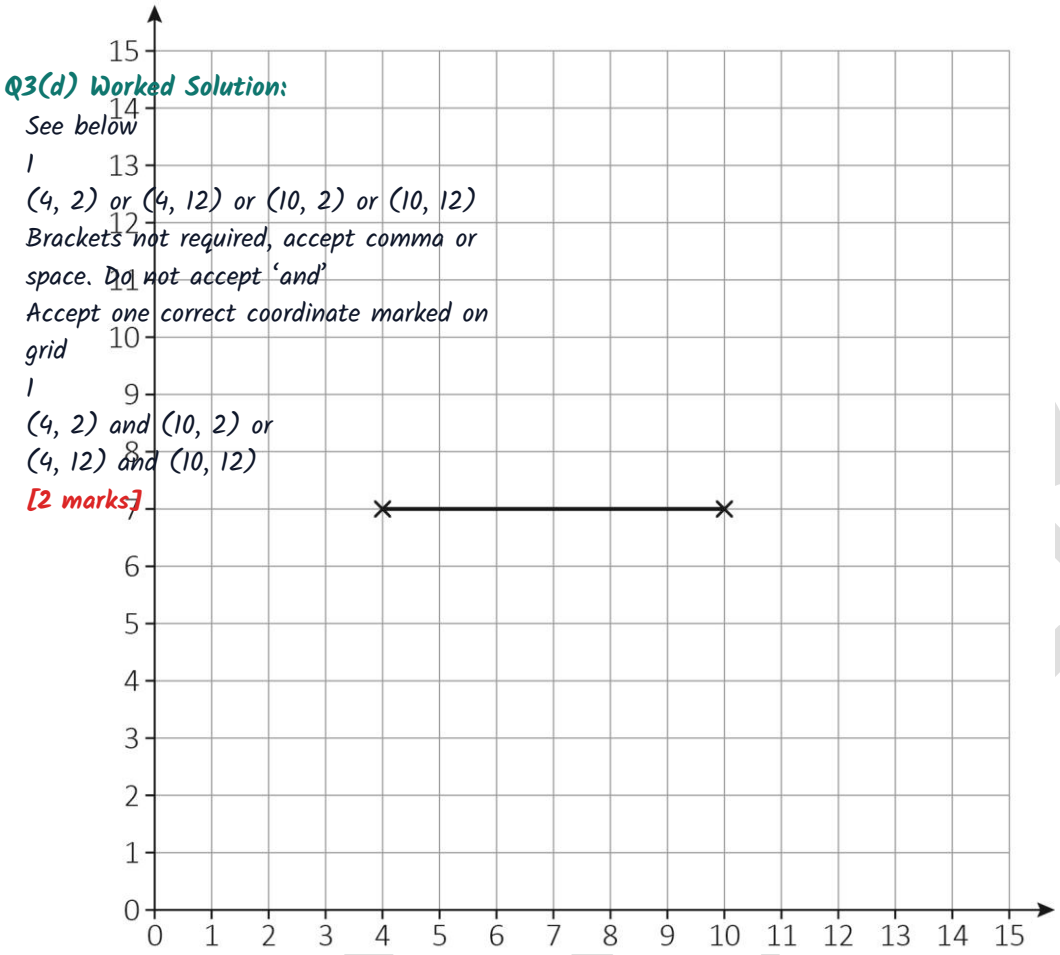
[2 marks]

Your answer:

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Examiner use only

3 (d) The coordinate grid shows the space for the car park:



Q3(d) Worked Solution:
See below
1
(4, 2) or (4, 12) or (10, 2) or (10, 12)
Brackets not required, accept comma or space. Do not accept 'and'
Accept one correct coordinate marked on grid
1
(4, 2) and (10, 2) or
(4, 12) and (10, 12)
[2 marks]

The car park is rectangular. It measures 120 m by 100 m

Each square on the grid represents 20 m by 20 m

Pavel draws two vertices and one edge of the car park area on the grid.

Write down possible coordinates for the other two vertices of the car park.

[2 marks]

Your answer:

Please turn over

Examiner use only

3 (e) The area where the festivalgoers will have their tents is 314 acres.

Pavel knows that for safety reasons each tent requires 27 m^2

He thinks that more than 50 000 tents can fit into this area.

Is he correct?

Show how you decide.

Use the conversion: 1 acre = 4046 m^2

[3 marks]

Q3(e) Worked Solution:

See below

8

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1

314 × 4046 or 1 270 444

1

Their 1 270 444 ÷ 27 or 47 053.48..

OR

Their 1 270 444 ÷ 50 000 or 25.4(...)

Implies 1st mark

[3 marks]

Your answer:

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Examiner use only

3 (f) The probability that a festivalgoer chosen at random has travelled by car is $\frac{14}{25}$

What is the probability that they **did not** travel by car?

Give your answer as a decimal.

[2 marks]

Q3(f) Worked Solution:

0.44

Award 2 marks if correct answer given

1

1 – 14

25

or 11

25

or 0.56

H27

1

[2 marks]

Your answer:

[Total marks: 15]

PAST PAPER

Please turn over

Examiner use only

Activity 4: College admissions

4 (a) Mel is an apprentice in a college admissions office.

She is organising interviews for new applicants.

The ratio of interviewers to applicants each day should be no higher than 2 : 15

There are 5 interviewers available on Monday.

Q4(a) Worked Solution:

37

Award 2 marks if correct answer given

1

5 ÷ 2 × 15 or 37.5

1

37

[2 marks]

What is the maximum number of applicants Mel should invite on Monday?

[2 marks]

Your answer:

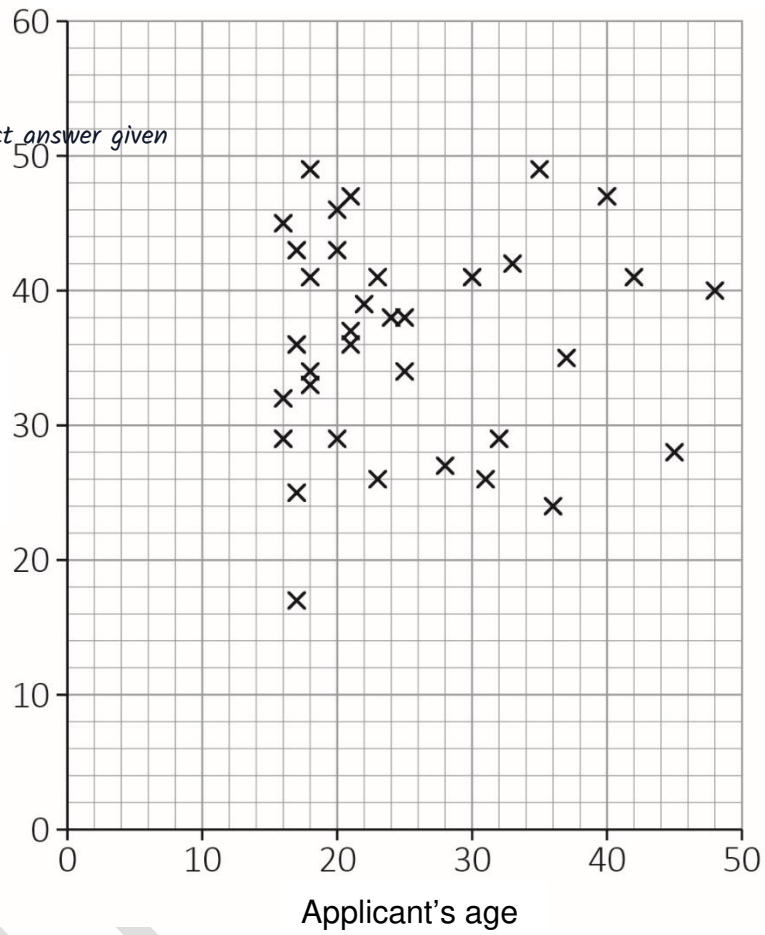
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Examiner use only

4 (b) All applicants who attend the interview complete an initial assessment.

Mel has this scatter diagram about the results of the initial assessment:

Initial assessment marks and age of applicants



Q4(b) Worked Solution:
5
12
Award 3 marks if correct answer given
9
Level 2 – P001491 v1.5
1
12
H28
1
5
[3 marks]

What fraction of applicants over the age of 26 scored more than 40 marks?

[3 marks]

Your answer:

Please turn over

Examiner use only

4 (c) Mel has information about applicants accepted on two new courses:

	Course A	Course B
Male applicant	22	71
Female applicant	59	48

Q4(c) Worked Solution:

What is the probability one of these applicants chosen at random is female and accepted on Course B?

Give your answer as a percentage.

[2 marks]

24(%)
Award 2 marks if correct answer given
1
48
200
OE or 0.24
H26
1
24(%)
H26
[2 marks]

Your answer: ____ %

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Examiner use only

4 (d) Mel needs to write a report about the income from tuition fees.

She has this information about tuition fees:

Course	Fee
A	£390
B	£650
C	£720
D	£460
E	£390
F	£980
G	£720
H	£980
J	£720

Mel writes in her report,

'The college has a £1.7 million income target this year.'

If 1476 students enrol onto courses and pay the modal tuition fee, the college will achieve more than $\frac{5}{8}$ of this target.

Is this statement correct?

Show how you decide.

[4 marks]

Your answer:

Please turn over

PAST PAPER

Question 4(d) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q4(d) Worked Solution:

See below
1
(£)720
Modal value identified
H23b
1
1 700 000 ÷ 8 × 5 or 1 062 500
Allow full method to find 5
8
of any value
[4 marks]

Examiner use only

4 (e) The conversion rate is the proportion of all applicants who go on to enrol on a course.

This proportion is expressed as a decimal.

Mel needs to compare the conversion rates for this year and last year.

Q4(e) Worked Solution:

Mel has this information about enrolments this year:

See below
1

- all applicants: 7680
- applicants who enrolled on a course: 4167

6 ÷ 11 or 0.5454.. or

Last year, 6 out of every 11 applicants enrolled on a course.

Is the conversion rate this year lower than the conversion rate last year?

Show how you decide.

[2 marks]

4167 ÷ 7680 or 0.5425.. OR
6 ÷ 11 × 7680 or 4189.09.. OR
4167 × 11 ÷ 6 or 7639.5
1
Yes AND 0.545(4..) and 0.542(5..) OR
Yes AND 54.5(4..%) and 54.2(5..%) OR
Yes AND 4189(.09..) OR
[2 marks]

Your answer:

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Examiner use only

4 (f) A new student takes out a student loan of £1450 for 2 years at a compound rate of 3% per annum.

Work out the amount of interest owed on this loan after 2 years.

[2 marks]

Q4(f) ~~Worked Solution:~~
(£)88.30 or (£)88.31
Award 2 marks if correct answer given
1
1450 × 1.032 − 1450 or 88.305
OE any full process to find the total interest
1
(£)88.30 or (£)88.31
2dp required
[2 marks]

Your answer: £

[Total marks: 15]

This is the end of the assessment.

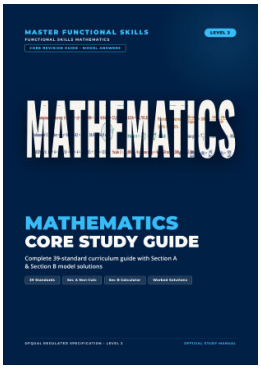
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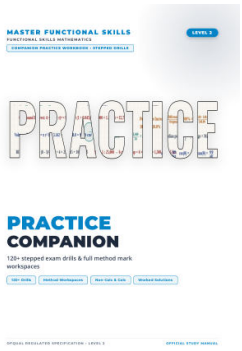


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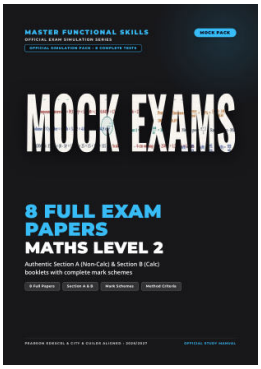


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